Statistical Analysis

MA222 Syllabus Section BL4 Fall 2026

Info

3 Credits; 2 lecture and 2 lab hours
Mon 11:00am - 1:00pm
Room: B302A

Instructor Information

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office: B831 Science and Math
office hours: M 1-3, W 11-12, R 12-1, or appointment

Description

Studies the principles and methods of statistical analysis including probability distributions, sampling distributions, error of estimate, significance tests, correlation and regression, chi-square, and ANOVA. Introduces the use of the computer to store, manipulate, and analyze data. Prerequisite(s): mathematic proficiency (see beginning of Mathematics section).

Outcomes

Upon completion of this course, students will be able to:

1. Summarize and graphically display data.
2. Compute and interpret measures of center and spread.
3. Work with probability distributions, including the normal distribution.
4. Use sampling distributions and describe the error of estimate.
5. Construct and interpret confidence intervals.
6. Carry out significance tests and interpret the results.
7. Compute correlation and fit a regression line.
8. Apply chi-square tests.
9. Apply analysis of variance (ANOVA).
10. Use the computer to store, manipulate, and analyze data.

Meeting Dates

This section meets in person every other Monday. The work in the weeks between meetings is done online.

1. Mon, Aug 31
2. Mon, Sep 14
3. Mon, Sep 28
4. Mon, Oct 12
5. Mon, Oct 26
6. Mon, Nov 9
7. Mon, Nov 23
8. Mon, Dec 7
9. Mon, Dec 14

Course Materials

Textbook

Intro Statistics is free and online. No other textbook is required.

Software

We will be using Google Spreadsheets or other free software for all work in this course. Since these are web-based applications there is NO OTHER SOFTWARE required for the course besides a web browser.

Topics

- Displaying and Summarizing Data
- Measures of Center and Spread
- Probability
- Probability Distributions
- The Normal Distribution
- Sampling Distributions and Error of Estimate
- Confidence Intervals
- Significance Tests
- Correlation and Regression
- Chi-Square Tests
- Analysis of Variance (ANOVA)

Evaluation

Your grade will come from these parts:

- Quizzes (88%)
- In Class/Online Work Credits (12%)

Each of these parts is described in more detail below

Quizzes

Your quiz grade will come from 4 quizzes, roughly covering 2 meetings of material each. The quizzes are 30 minutes each and are usually 5 or 6 questions each. These quizzes are with no notes, no internet, no phone, no software, no AI tools. Pen and paper and calculator only. They are some multiple choice, some short answer, some true false.

In Class/Online Work Credits (1-3 per class)

These are credits you obtain for demonstrating you have completed assigned problems. There is no homework in this course.

Some of these will come from in class assignments that are done during our in person meetings and you show as you complete them. The rest come from the online work assigned for the weeks between meetings. You will earn a single credit for each successful assignment completion. I decide if the work is complete enough to receive credit for.

The in class work can only be turned in by showing it to me during a meeting. There is not a way to turn that work in by emailing it to me or submitting it in Brightspace. You show it to me and I mark it as complete when you are in class. You must be in attendance to earn those credits.

The online work is turned in online by the date it is due.

You may miss up to 1 meeting and still show me the in class work from that meeting at the next meeting. But you may not do this more than once.

Midterm, Final Exam

There is NO MIDTERM and NO FINAL EXAM in this course.

AI Policy

Generally all assignments use AI in some form. But you must only use the AI in the way the assignment calls for. Every other use of AI outside of the specific assignment methods or techniques is not allowed.

Institutional Policies and Resources

The academic honor code, withdrawal procedures, emergency preparedness, student conduct policy, and academic and technical support resources all apply to this course:

Institutional Policies and Resources